Physics - Graduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. When the temperature of a strip of iron is increased, the length of the strip
   (a) increases at first, then decreases.
   (b) decreases in width as it gets longer.
   (c) increases exponentially.
   (d) remains the same.
   (e) may increase and may decrease.

2. Using the speed of sound at sea level as approximately 750 miles per hour, what is the velocity of a jet plane traveling at Mach Number 2.2?
   (a) 1900 mph
   (b) 1750 mph
   (c) 1320 mph
   (d) 1200 mph
   (e) 1800 mph

3. What is the radiant intensity of aLambertianemitter which projects 50mWinto a hemisphere?
   (a) .00995 (W /sr)
   (b) .01095 (W/sr)
   (c) .01295 (W/sr)
   (d) .00695 (W /sr)
   (e) .00295 (W/sr)

4. Assume that half of the mass of a 62-kg person consists of protons. If the half-life of the proton is 10^33 years, calculate the number of proton decays per day from the body.
   (a) 3.5e-08
   (b) 4.8e-05
   (c) 1.2e-06
   (d) 6.3e-04
   (e) 8.2e-06

5. When food is exposed to gamma radiation, the food
   (a) should be avoided
   (b) becomes slightly radioactive

(c) will become carcinogenic
(d) becomes inedible
(e) will glow in the dark

True / False (10 marks)

Answer True or False

6. True or False: The current in the connecting wire of an LED lamp is half as much as the current f
7. True or False: An object starting from rest with an acceleration of 16 ft/s for 3 seconds will attain
8. True or False: An ocular lens with a focal length of 2.5 cm will yield an overall magnification of 10
9. True or False: When a voltage of 120 volts is applied to a light bulb with a resistance of 240 ohms
10. True or False: The focal length of a diverging mirror is positive when the image is formed behind

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of alpha decay in uranium-238, including the resulting transformation of the nucleus and the implications for nuclear stability and radioactive decay chains.
12. Calculate the power density of a laser with a given power and beam diameter, considering a specific loss in the focusing system. Discuss the implications of your findings.

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